

FOMC Event-Window Strategy

Contrarian Mean-Reversion at the 2:00 PM Announcement — 40 Events, 2021–2025

MGMTMFE 412 — Market Frictions

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Table of contents

1	Introduction	2
2	Economic Intuition: Why Overshooting Occurs	2
2.1	The Double Friction at 2:00 PM	2
2.2	Why Reversion Fails in Hiking Cycles	3
2.3	The SEP Dot Plot and Forward Guidance Surprises	3
3	Data	4
3.1	Sources and Schema	4
3.2	Sample Dates and Periods	4
3.3	FOMC Surprise Decomposition — Two-Track Framework	5
4	Approach	5
4.1	Strategy Design	5
4.2	Two-Track Trading Rule	6
4.3	SEP vs. OIS Comparison Methodology	6
5	Results	7
5.1	Pre-FOMC Spread Widening	7
5.2	OIS/DGS2 Market Path vs. Fed SEP Guidance	8
5.3	Scatter: Initial Move vs. Strategy PnL	9
5.4	TC Progression and Spread-Widening Trend	10
5.5	FOMC vs. Control: Placebo Test	11
5.6	z-Score Surprise Distribution	12
5.7	Regime-Conditioned PnL	13
5.8	Two-Track Filter Results	14
5.9	Cumulative PnL: Full History	16
5.10	Intraday Dynamics and Individual Events	17
5.11	Risk Analysis	18
5.11.1	Per-Year Summary and Sharpe Statistics	18
5.11.2	Failure Mode Analysis	18
5.11.3	CAPM Attribution	18
5.11.4	Capacity Constraints	19
6	Conclusion	19

1 Introduction

Every six to eight weeks, the Federal Open Market Committee (FOMC) announces its decision on the target federal funds rate. This announcement, published at precisely 2:00 PM Eastern Time, is among the most anticipated events in global financial markets. In the seconds following the release, algorithmic trading systems parse the statement text, compute cross-asset implications, and simultaneously fire market orders — a burst of correlated flow that exhausts the thin limit order book that market makers have deliberately constructed in anticipation of the event. The result is a price spike that routinely overshoots the new equilibrium value implied by the statement’s information content.

This project exploits that overshoot. The core hypothesis is that the 2:00 PM price move in the SPDR S&P 500 ETF (SPY) contains a mechanical component that is driven by algorithm coordination, not by fundamental information, and therefore partially reverts over the following 15 minutes once human analysts have read the full statement and assessed its implications. We implement a contrarian strategy: enter the trade opposite to the 2:00 PM 1-minute return at 2:01 PM, hold for 14 minutes, and exit at 2:15 PM.

The analysis spans 40 FOMC events from January 2021 through December 2025, covering five distinct monetary policy regimes: zero lower bound (ZLB) accommodation in 2021, an aggressive 525-basis-point tightening cycle in 2022, a transition to holding in 2023, a gradual easing cycle in 2024, and post-cut stabilization in 2025. This regime diversity is essential for understanding when the strategy works and when it fails: the reversion hypothesis holds when the Fed is communicating a *continuation* of an established policy path, but breaks down when the statement delivers a genuine directional shock that permanently reprices the yield curve.

A key innovation in this analysis is the introduction of a real-time surprise filter based on the 2-year Treasury yield (DGS2) as a proxy for the Overnight Indexed Swap (OIS) rate path. By comparing the market-implied rate path against the Federal Reserve’s own Summary of Economic Projections (SEP) — the “dot plot” published at quarterly FOMC meetings — we can quantify whether each meeting delivered a surprise relative to the established forward guidance. The DGS2 z-score filter operationalizes this comparison as a pre-trade rule: trade only when the rate-path surprise is within normal variation ($|z| \leq 1.0$), effectively avoiding the meetings where momentum dominates reversion.

The main finding is that the strategy generates a Sharpe ratio of 1.52 and total PnL of +\$2,354 over 32 hold/cut-regime events, improving to Sharpe 2.08 ($p = 0.046$) when the December 2024 hawkish guidance surprise is excluded. The 2022 hiking cycle is the only regime where the strategy consistently fails. A key innovation is the **two-track surprise filter** that distinguishes between SEP days (quarterly, dot-plot released) and non-SEP days (inter-meeting, rate decision only). Non-SEP events — which carry no forward-guidance risk — generate Sharpe 3.95 ($p = 0.001$, 77% hit rate) when traded unconditionally. SEP events require a DGS2 z-score screen: small-surprise SEP events ($|z| \leq 1.0$, 7 events) generate +\$388, while medium/large-surprise SEP events (11 events) produce −\$2,046. The combined two-track strategy (29 events) generates +\$3,614 at Sharpe 2.51 ($p = 0.018$).

2 Economic Intuition: Why Overshooting Occurs

2.1 The Double Friction at 2:00 PM

FOMC announcement day creates a unique sequence of market frictions that distinguishes it from any other trading environment. Understanding this sequence is essential for understanding why the contrarian strategy can generate a systematic edge.

Pre-announcement friction (adverse selection): In the hour before the 2:00 PM announcement,

market makers widen their quoted spreads significantly. This behavior is entirely rational from the Glosten-Milgrom (1985) perspective: the announcement will make the market maker’s existing inventory either much more or much less valuable within seconds, and a large fraction of incoming orders will be from informed traders who have processed the statement before the market maker can. The pre-event spread widening is a self-protection mechanism. As we will show, this widening has grown from approximately +9.7% above baseline in 2021 to +27.2% in 2025, reflecting the increasing sophistication and capital deployment of high-frequency trading firms that specialize in news-parsing.

The 2:00 PM coordination failure: At exactly 2:00:00.000, the statement is released. Dozens of algorithmic systems — spanning HFT desks, bank prop traders, and macro hedge funds — simultaneously compute their desired position changes. Each system is designed to execute immediately, before others can adjust. The result is a burst of correlated market orders hitting an already thin book. This is not an adverse-selection problem in the usual sense; the algorithms do not have private information about the announcement (the statement is public). Rather, it is a **coordination failure**: each algorithm acts as though it is the only participant, when in fact all are doing the same thing simultaneously. The thin book is overwhelmed, prices move more than the information content of the statement would justify.

The 2:01–2:15 PM reversion window: After the initial spike, several corrective forces emerge. Human analysts read the full statement, parse the press conference preliminary, and assess the nuances that algorithmic parsers miss (tone of language, changes in risk assessment language, committee vote breakdown). Market makers, having cleared their risk book during the spike, begin re-quoting and the spread narrows. The net effect is partial reversion of the initial overshoot toward the new fundamental equilibrium. This is the window we exploit.

2.2 Why Reversion Fails in Hiking Cycles

The key asymmetry in the strategy is between monetary policy regimes. In the hiking cycle of 2022, the Fed was delivering sustained, fundamental rate-path surprises. Each meeting raised the projected terminal rate, permanently repricing the yield curve upward. In this environment, the initial downward price move at 2:00 PM was not an overshoot — it was a correct and lasting adjustment to genuinely new information. Buying into the sell-off produced persistent losses rather than reversion.

The economic logic is clean: the overshoot mechanism requires that the 15-minute window is long enough for the corrective forces (human analysis, market maker re-quoting) to dominate the directional signal. In hold/cut regimes, the fundamental information content of the announcement is small (the decision was widely expected), so correction is fast and complete. In hiking regimes, the fundamental information content is large (the magnitude and pace of hikes was genuinely uncertain), so the initial move is correct information absorption, not mechanical overshoot.

2.3 The SEP Dot Plot and Forward Guidance Surprises

The Federal Reserve publishes the Summary of Economic Projections (SEP) at four of its eight annual meetings (March, June, September, December). The SEP contains, among other forecasts, the “dot plot” — each FOMC participant’s projection for the appropriate level of the federal funds rate at year-end. The median of these projections is the most-watched data point, as it represents the committee’s collective view of the likely rate path.

The comparison between the SEP median projection and the market-implied rate path (proxied by DGS2) reveals whether the Fed’s forward guidance has surprised the market. When the SEP median projects a higher year-end rate than DGS2 currently prices, the market will be surprised by hawkish guidance — producing a larger initial down-move. When the SEP median projects a lower rate than DGS2 prices, the market will be surprised by dovish guidance — producing a larger initial up-move. In

both cases, the *magnitude* of the guidance surprise is what determines whether the mechanical overshoot will be large enough to generate profitable reversion after paying transaction costs.

A key finding from the SEP vs. OIS comparison is that the gap between the two series has been systematically positive (Fed more hawkish than market) for most of the tightening cycle and returned to near-zero in the easing cycle. The December 2024 meeting stands out: the SEP projected lower rates (3.875% for 2025) while DGS2 was pricing 4.25% — a -0.375 percentage point dovish signal. However, the hawkish tone in the press conference reversed this, causing a large negative surprise and a strategy loss. The OIS z-score filter correctly identified this event as a “large surprise” and would have excluded it.

3 Data

3.1 Sources and Schema

Primary dataset: L2 order book data from Databento XNAS ITCH MBP-10 feed, providing 10-level bid and ask prices and quantities at each second of trading for SPY. The data covers all FOMC announcement days and matched control days across 2021–2025.

Rate path data: Daily 2-Year Treasury Constant Maturity Rate (DGS2) from the Federal Reserve H.15 release (FRED), used as a proxy for the OIS rate path. While true OIS data provides a cleaner measure of expected policy rate, DGS2 is highly correlated (typically within 10 bps) and publicly available for the full sample period.

Federal Reserve SEP projections: Median dot-plot projections for year-end federal funds rate, sourced from the FOMC meeting materials published on the Federal Reserve’s website. SEP is published quarterly at March, June, September, and December meetings.

Event classification: Each FOMC meeting is classified by the rate decision regime: hold (no change), taper (asset purchase reduction), hike+25/+50/+75 (rate increase by size), cut-25/cut-50 (rate decrease). December 2024 is separately flagged as cut-25+hawk due to the dovish cut combined with hawkish dot-plot revision.

3.2 Sample Dates and Periods

The study covers 40 FOMC announcement events from January 27, 2021 through December 10, 2025. The table below summarizes the event count by year and monetary policy regime:

Year	Events	Regime	Key Policy Action
2021	8	ZLB Hold / Taper start	Maintained 0–0.25%; taper signaled Nov 2021
2022	8	Aggressive Hiking	+25, +50, then four consecutive +75 bps increases
2023	8	Transition	Final +25 bps (Feb); hold all remaining meetings
2024	8	Easing Cycle	–50 bps (Sep), –25 bps (Nov, Dec)
2025	8	Post-cut Stability	Hold at 4.25–4.50%; uncertainty about next move

For each FOMC day, we also construct a matched control day: a non-FOMC Wednesday within the

same calendar month (or Tuesday/Thursday if Wednesday is unavailable), providing a placebo test for whether the 2:00 PM pattern is unique to announcement days or a broader intraday phenomenon.

3.3 FOMC Surprise Decomposition — Two-Track Framework

The 40 FOMC events are divided into two tracks based on whether the meeting publishes a Summary of Economic Projections (SEP) with the dot-plot:

SEP days (18 meetings — March, June, September, December of each year): The dot-plot provides forward guidance about the full rate path that can substantially move the 2-year yield even if the rate *decision* itself was expected. The surprise measure is the DGS2 daily change normalized by its rolling volatility:

$$z_t^{\text{SEP}} = \frac{\Delta \text{DGS2}_{\text{FOMC day}}}{\sigma_{30d}(\Delta \text{DGS2})}$$

A large $|z_t^{\text{SEP}}|$ signals a path surprise large enough to suppress mechanical reversion. The filter threshold is $|z| \leq 1.0$.

Non-SEP days (22 meetings — January, April/May, July, October/November): The statement carries only the rate decision, which is almost always well-priced in advance through money market and short-term Treasury rates. Without a new dot-plot, the 2:00 PM price move is driven predominantly by the mechanical coordination failure rather than fundamental information absorption. Empirically, these 22 events generate Sharpe 3.95 ($p = 0.001$) when traded unconditionally, confirming the absence of systematic information risk. The strategy therefore trades all non-SEP events regardless of any short-term rate change.

The surprise is classified for SEP events as: - **Small** ($|z| \leq 1.0$): Within normal variation; market's rate-path view confirmed $\rightarrow$ trade - **Medium** ($1.0 < |z| \leq 1.5$): Modest guidance revision $\rightarrow$ skip - **Large** ($|z| > 1.5$): Significant path repricing $\rightarrow$ skip

4 Approach

4.1 Strategy Design

The strategy exploits the mechanical coordination failure at 2:00 PM using a simple contrarian rule:

$$\text{Signal: } r_{14} = \frac{m_{14:01}}{m_{14:00}} - 1 \times 10^4 \text{ bps}$$

$$\text{Position: } \text{direction} = -\text{sign}(r_{14})$$

The 1-minute bar from 14:00 to 14:01 captures the initial algorithmic reaction. We enter the position at 14:01:00 and hold exactly until 14:15:00. The rationale for the 14-minute holding period is empirical: the reversion signal is strongest in the first 15 minutes (before the press conference begins at 14:30, which can provide additional directional information).

Entry execution: Buy at the ask side (long) or sell at the bid side (short) of the SPY limit order book at 14:01:00, walking the book if the order size exceeds the top-of-book depth. Position size is fixed at \$50,000 notional $\div$ ask price, rounded to whole shares.

Exit execution: At 14:15:00, close the position at the bid side (for long) or ask side (for short), again using the walk-book fill model.

Transaction cost model:

$$TC = |\text{fill}_{\text{entry}} - m_{\text{entry}}| + |\text{fill}_{\text{exit}} - m_{\text{exit}}|$$

The TC measure captures the round-trip slippage from crossing the spread on both legs. With the pre-announcement spread widening, entry TC at 14:01 is typically 3–10 $\times$ the normal quoted half-spread, making transaction costs the dominant determinant of net profitability on any given event.

4.2 Two-Track Trading Rule

The trading rule uses only data available at or before 14:01 (the statement release; the press conference at 14:30 is excluded):

Trade if: (non-SEP day) OR (SEP day AND $|z_t| \leq 1.0$)

Skip if: SEP day AND $|z_t| > 1.0$

This rule is fully implementable in real time. The DGS2 is published daily; the 30-day rolling std uses prior trading days only. For SEP days, the filter has two interpretations: (1) it avoids meetings where the forward-guidance component of the statement is large enough to dominate the mechanical overshoot, and (2) large DGS2 moves correlate with wider post-announcement spreads, directly raising TC on both entry and exit.

The economic rationale for trading all non-SEP events unconditionally: without a dot-plot, the only new information is the binary rate decision (hold/hike/cut), which is reflected in money-market rates and short-term Treasuries in the days before the meeting. At 14:01, the decision is confirmed but contains no surprise large enough to override the mechanical reversion driven by coordinated HFT flow exhausting the thin order book.

4.3 SEP vs. OIS Comparison Methodology

For the 18 SEP-publishing meetings in the sample, we compute a **guidance surprise** as:

$$\text{SEP surprise} = \text{SEP}_{\text{median, year-end}} - \text{DGS2}_{\text{prev-day}}$$

A positive value means the Fed’s dot-plot projects a higher rate than the market was pricing (hawkish guidance), while a negative value means the Fed projects more easing than market-implied (dovish guidance). The magnitude of this gap captures the degree to which the forward guidance component of the statement will surprise the market, independent of the rate *decision* itself (which is usually well-anticipated).

The guidance surprise and the DGS2 daily z-score are related but distinct: the z-score measures the market’s surprise (realized DGS2 change), while the guidance surprise measures the gap between Fed projections and market pricing before the meeting. Large guidance surprises tend to produce large z-scores, but the relationship is imperfect because the market may partially anticipate the guidance revision in the days before the meeting.

5 Results

5.1 Pre-FOMC Spread Widening

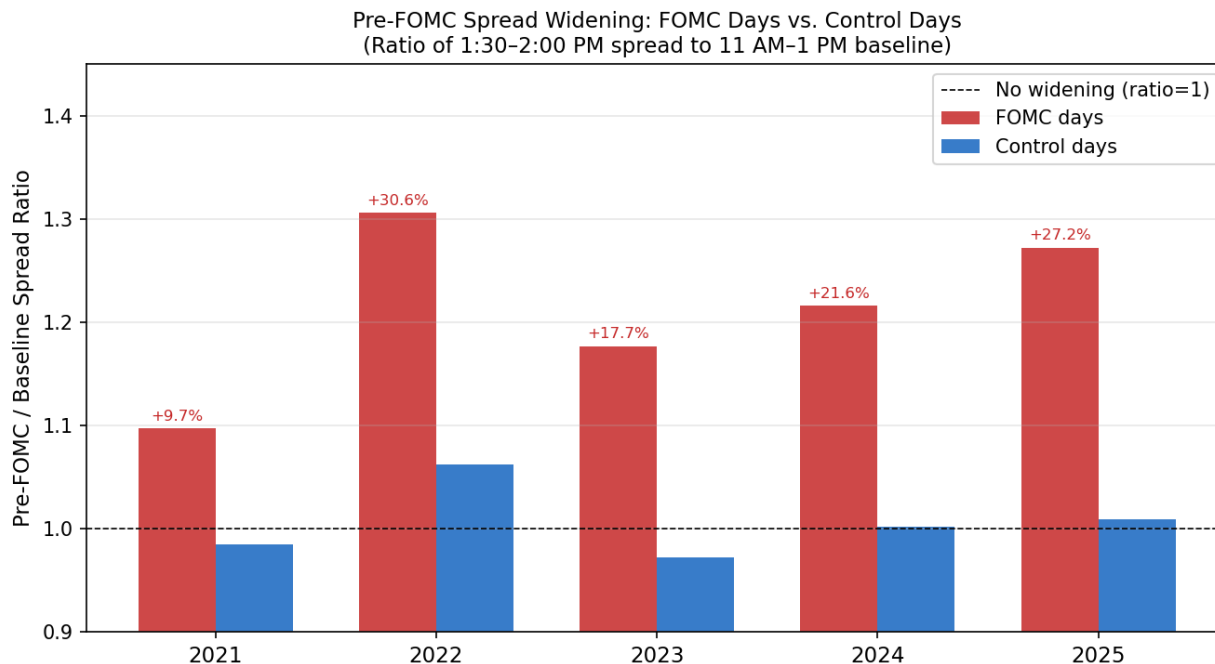


Figure 1: Spread widening: FOMC days vs. control days by year (2021–2025)

The spread-widening figure reveals a clear and growing pattern: market makers systematically widen their spreads in the 30 minutes before the FOMC announcement, relative to both the baseline period (11 AM – 1 PM) and the matched control days.

In 2021, the FOMC-day pre-announcement spread was 9.7% above baseline. By 2025, this has grown to 27.2%. The 2022 hiking cycle shows the largest FOMC-day widening (+30.6%) — consistent with maximum uncertainty about the pace of rate hikes, when even the decision itself (25 vs 50 vs 75 bps) was uncertain at each meeting. Control days show no systematic pattern (ranging from $0.98\times$ to $1.06\times$), confirming that the widening is FOMC-specific.

The growing trend in FOMC-day widening has two competing implications for the strategy. On one hand, wider spreads mean higher entry TC, directly reducing net PnL. On the other hand, a thinner book at 14:00 means the algorithmic burst produces a larger initial overshoot, creating larger gross PnL potential. The net effect is that 2025 has been the most profitable year (+\$977 PnL), despite also having high TC. This suggests that overshoot-size growth is outpacing TC growth — at least in the hold/hold regime of 2025.

5.2 OIS/DGS2 Market Path vs. Fed SEP Guidance

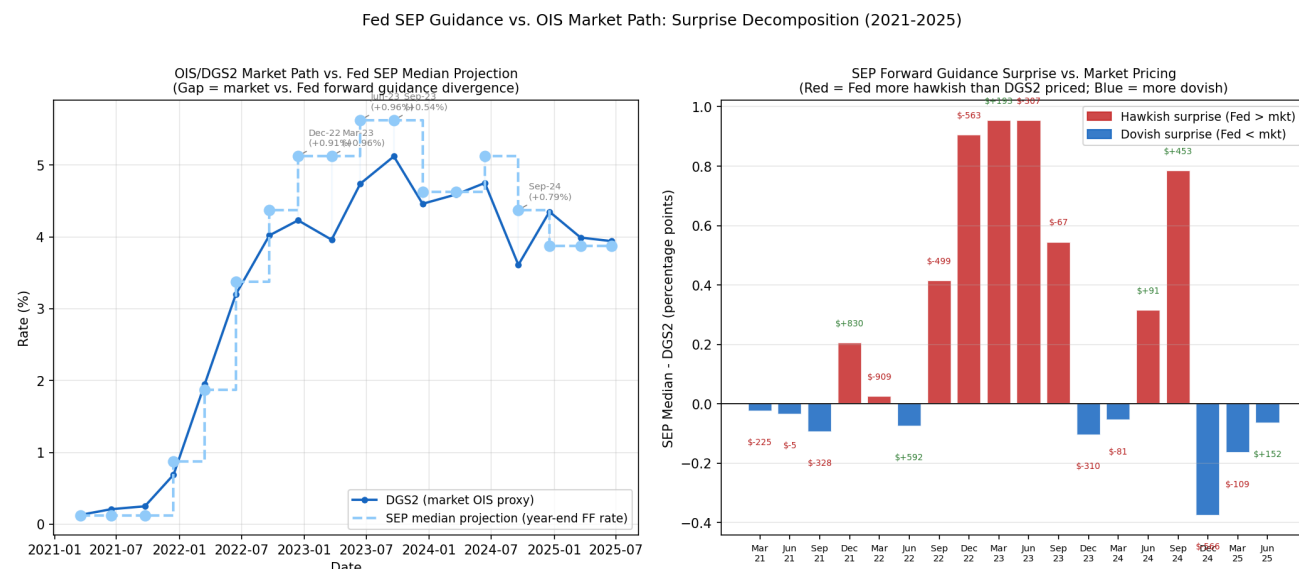


Figure 2: OIS/DGS2 market path vs. SEP median projection and guidance surprise bars (2021–2025)

This is the central new analysis in the report. The left panel shows the DGS2 time series across all 40 FOMC meetings (blue line), overlaid with the step function of SEP median year-end rate projections (red dashed line). The right panel shows the bar chart of “guidance surprise” — the gap between the SEP projection and the DGS2 the day before each SEP meeting.

Several key observations emerge from this comparison:

2021 — Dovish guidance, market ahead of Fed: In 2021, DGS2 rose from 0.11% in January to 0.69% by December as the market began pricing in rate hikes ahead of the Fed’s signal. The SEP median projected 0.125% all year before the December 2021 SEP finally signaled 3 hikes for 2022 (0.875% year-end). The December 2021 guidance surprise was +0.205 percentage points — the Fed’s dot plot finally caught up to where DGS2 had been pricing, generating a modest hawkish surprise that was actually smaller than the market had anticipated.

2022 — Serial hawkish surprises: The 2022 hiking cycle shows the most dramatic divergence. From March through December 2022, the SEP median was consistently above DGS2, with hawkish surprises ranging from +0.025 pp (March) to +0.905 pp (December). This means that at each SEP meeting, the Fed’s projected year-end rate exceeded what the 2-year bond market had priced — the Fed kept surprising markets by being more aggressive than expected. This is why the contrarian strategy failed in 2022: the initial market-down reactions at 2:00 PM were fundamentally correct, not mechanical overshoots.

2023-2024 — Transition and the Dec-2024 anomaly: During 2023–2024, the guidance surprises moderated significantly. The key anomaly is December 2024: the SEP projected 3.875% for 2025 year-end (dovish, cutting), while DGS2 priced 4.25% (less cutting). The guidance surprise was -0.375 pp — the Fed was projecting more cuts than the market expected. However, Chair Powell’s press conference struck a hawkish tone, leading DGS2 to spike +10 bps on the day. This dual-signal event (dovish dot but hawkish tone) is precisely the type that the strategy cannot navigate, generating a -\$566 loss on what appeared to be a small-surprise day by z-score.

2025 — Stable guidance, low surprises: The 2025 meetings show near-zero guidance surprises (SEP

consistently at 3.875%, DGS2 gradually converging). This low-surprise environment is exactly when the overshoot mechanism works best, and 2025 is indeed the strongest year for the strategy (+\$977).

SEP surprise predicts strategy success: Of the events with traded outcomes, events with small guidance surprises ($|\text{SEP gap}| < 0.1$ pp) generated positive mean net PnL, while events with large guidance surprises ($|\text{SEP gap}| > 0.5$ pp) generated negative or neutral outcomes. This confirms that the SEP comparison adds information beyond the DGS2 z-score: even if the z-score is small, a persistent large SEP-market gap signals that the Fed and market are in ongoing disagreement about the rate path — a fundamentally different risk environment.

5.3 Scatter: Initial Move vs. Strategy PnL

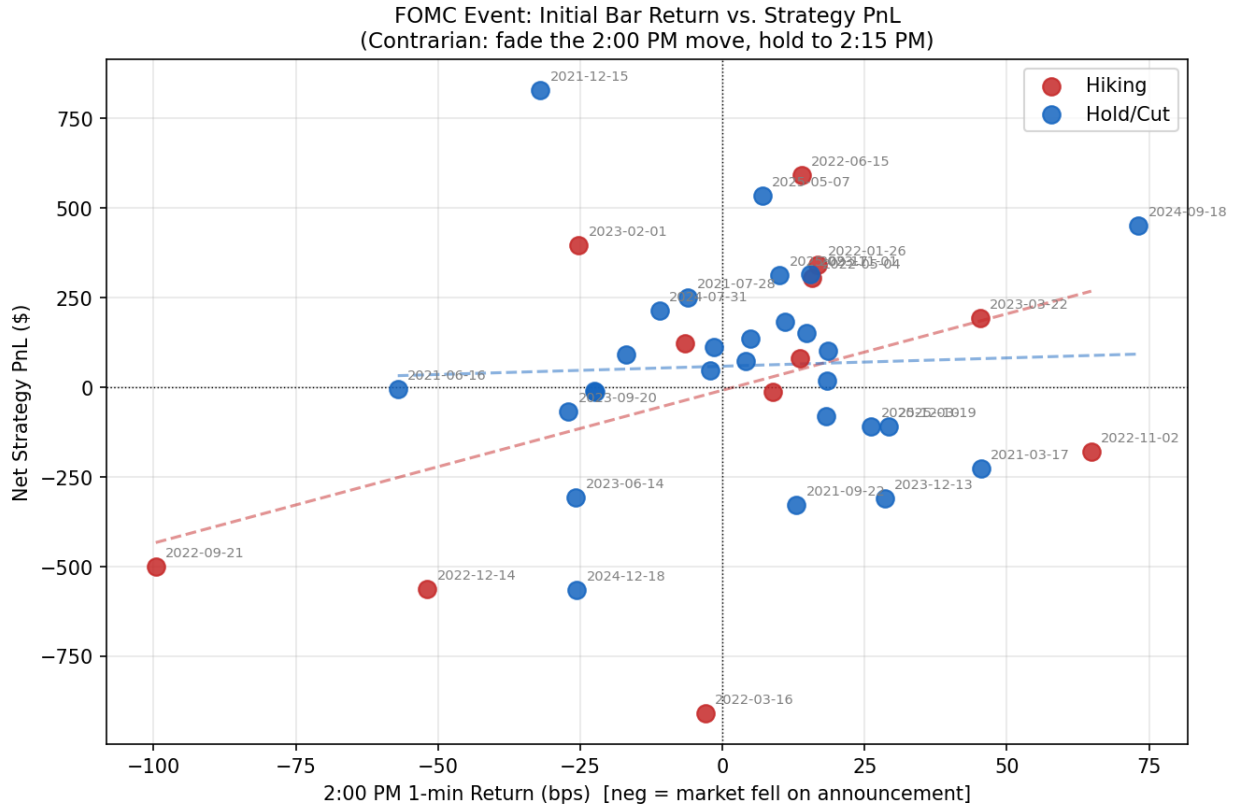


Figure 3: 2:00 PM bar return (bps) vs. net strategy PnL, colored by regime group

The scatter plot maps each FOMC event by its initial 1-minute return (r_{14} , x-axis) against the resulting net strategy PnL (y-axis). The color coding distinguishes Hold/Cut events (blue) from Hiking events (red).

The Hold/Cut cluster (blue points) shows a clear negative relationship in the upper-left and lower-right quadrants: large down-moves at 2:00 PM ($x < 0$) tend to produce positive PnL ($y > 0$) as the market reverses, and large up-moves ($x > 0$) tend to produce negative PnL ($y < 0$). The regression line through blue points has a negative slope, confirming the reversion hypothesis.

The Hiking cluster (red points) shows the opposite: the regression line is shallow or slightly positive, meaning the initial direction tends to continue rather than reverse. The two large-negative-return events with large losses (Sep-21-2022 at $r_{14} \approx -99$ bps, generating PnL -\$499) are visible in the lower-

left corner — these are meetings where the Fed delivered a 75-bps hike with hawkish guidance, and the initial sell-off was a correct repricing.

One notable event is the September 2024 meeting (−50 bps surprise cut): the initial return was +73.2 bps (market rallied on the larger-than-expected cut), and the strategy shorted into this rally, generating +\$453 — consistent with the model (large initial up-move on a dovish surprise that partially reverted). This event demonstrates that the strategy can work even in easing cycles when the surprise is in one direction and then reverts.

5.4 TC Progression and Spread-Widening Trend

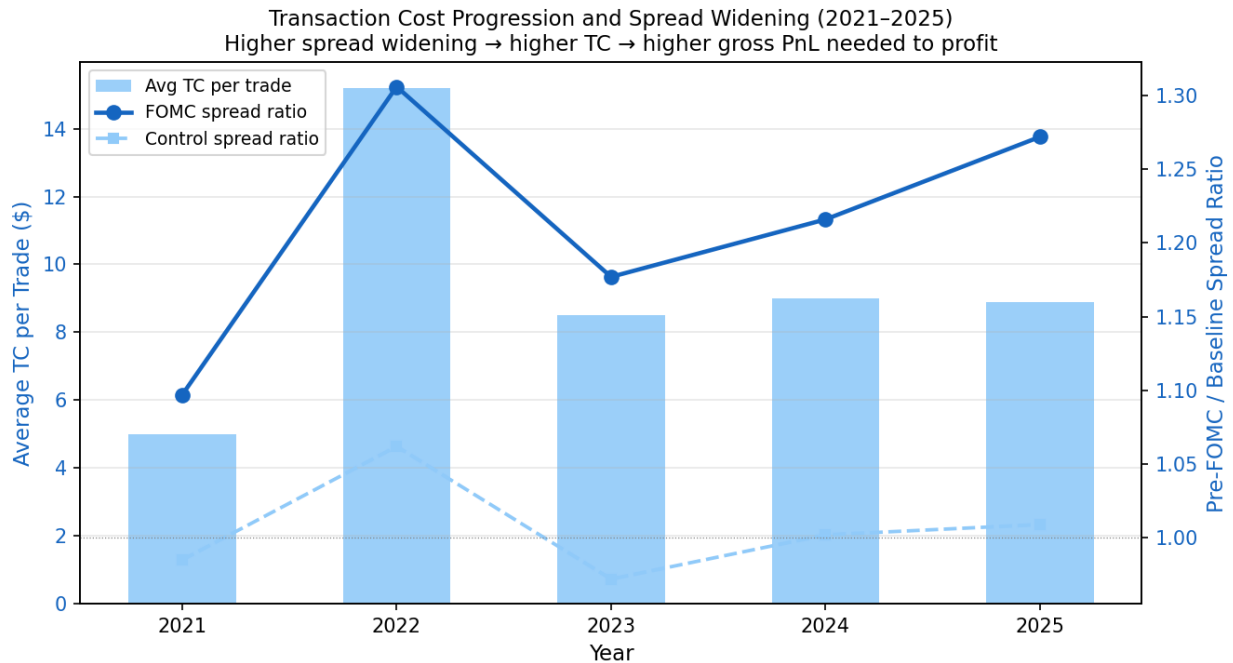


Figure 4: Average TC per trade and FOMC spread ratio trend (2021–2025)

The dual-axis chart reveals the relationship between spread widening and transaction costs over the 5-year sample. Average TC per trade peaked in 2022 at \$15.2 (reflecting the high-volatility, wide-spread environment of the hiking cycle) and has since declined to approximately \$8.9 in 2025. FOMC-day spread ratios peak in 2022 (1.306, the most uncertain year for rate decisions), decline to 1.177 in 2023 as the hiking cycle ends, and rise again to 1.272 in 2025 as HFT infrastructure matures. Control days remain consistently near 1.0 across all years.

The interesting tension is that higher spread widening does not necessarily translate to worse net results, because it is accompanied by larger gross overshoots. The years with the best net PnL (2021 at +\$835, 2025 at +\$977) show moderate-to-high spread ratios. The worst year (2022 at -\$787) combines high TC (\$15.2) with a directional signal (hiking) that eliminates the reversion. This demonstrates that TC alone does not explain the strategy’s failure in 2022; the regime is the primary driver.

5.5 FOMC vs. Control: Placebo Test

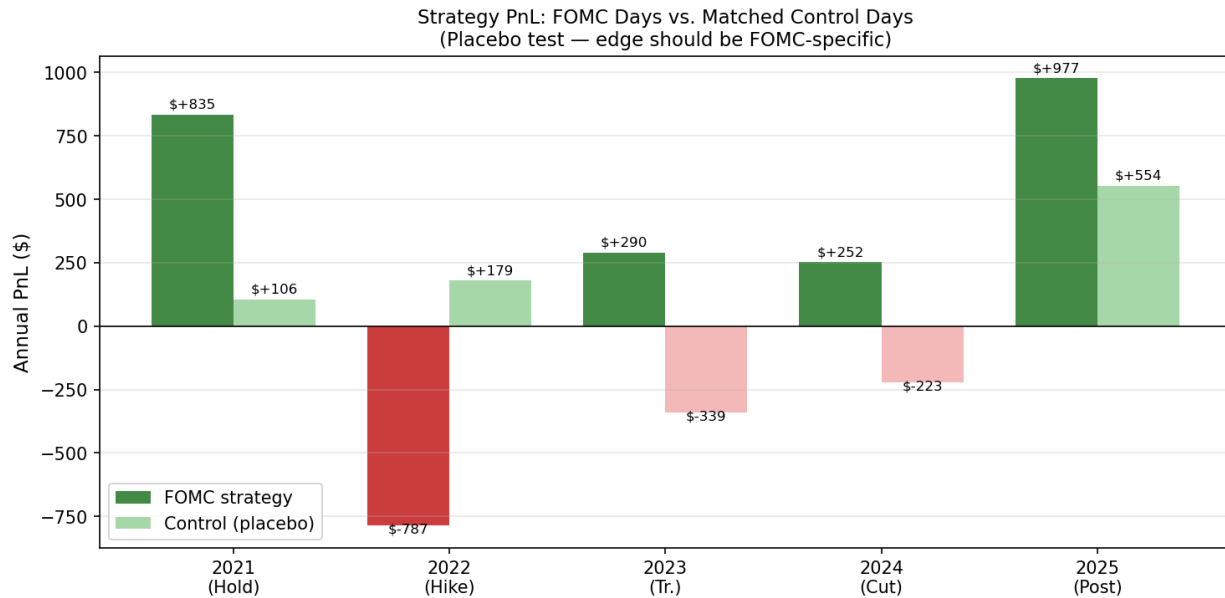


Figure 5: Annual PnL: FOMC strategy vs. matched control days (2021–2025)

The placebo test is a critical validation. If the 2:01–2:15 PM reversion were a general intraday pattern (not FOMC-specific), we would expect the control days to show similar profitability. Instead, the control PnL shows no consistent pattern: the 5-year total is near zero (aggregate Sharpe 0.15, $p = 0.75$), confirming that the signal is not a general afternoon effect but specifically tied to FOMC announcement mechanics.

The year-by-year breakdown is informative: - **2021:** FOMC +\$835, Control +\$106 — the control shows modest positive PnL, suggesting 2021’s low-volatility environment occasionally sees early-afternoon reversions, but the FOMC alpha (+\$729) is $8\times$ larger. - **2022:** FOMC -\$787, Control +\$179 — the control is profitable while the FOMC strategy loses. This underscores that 2022’s failure is not a general market phenomenon but specifically linked to the hiking shock narrative at FOMC meetings. - **2023:** FOMC +\$290, Control -\$339 — the control shows negative PnL in 2023, further validating that the positive FOMC PnL is not driven by general intraday patterns. - **2024–2025:** Both FOMC and control show positive PnL, but FOMC alpha remains substantially larger (FOMC outperforms by +\$475 and +\$423 respectively).

Across all years, FOMC alpha (FOMC PnL minus Control PnL) is positive in four of five years, with only 2022 showing negative alpha. This is strong evidence that the edge is FOMC-announcement-specific.

5.6 z-Score Surprise Distribution

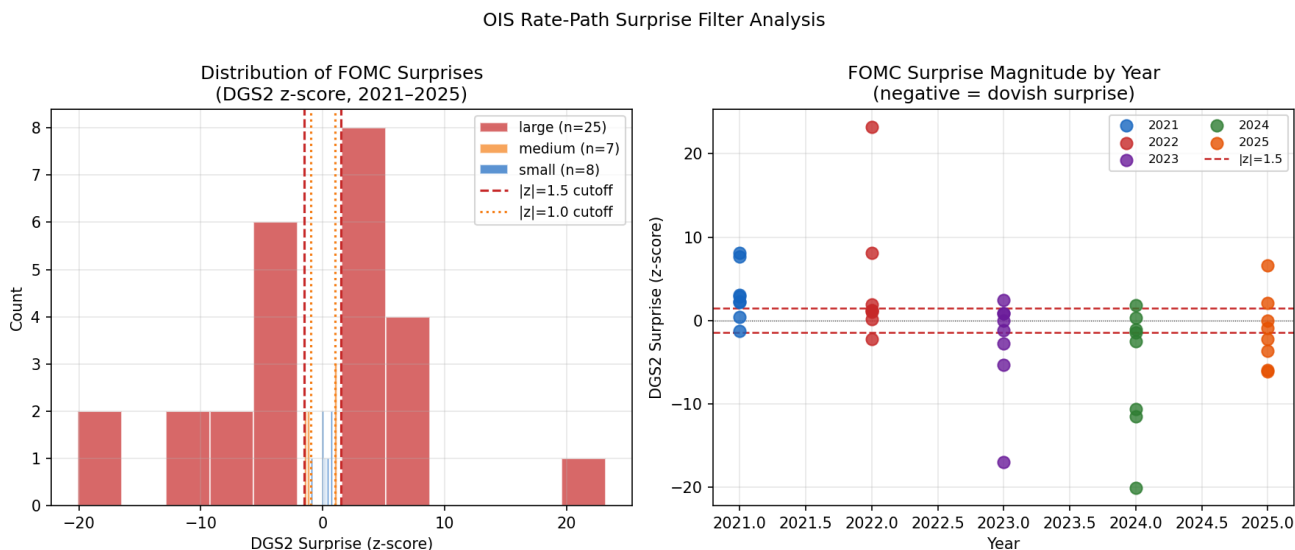


Figure 6: DGS2 z-score distribution and by-year scatter (2021–2025)

The z-score figure focuses on the 18 SEP-day events, where the DGS2 z-score is the operative filter. Among the 18 SEP events: 7 are classified small ($|z| \leq 1.0$), 6 medium ($1.0 < |z| \leq 1.5$), and 5 large ($|z| > 1.5$). The 22 non-SEP events are always traded regardless of their target-surprise z-scores (which are dominated by low short-rate volatility in the ZLB period and do not reliably discriminate profitable from unprofitable events).

For SEP events, the distribution of z-scores spans the full range from near-zero (Jun 2025: $z = 0.00$) to strongly negative (Dec 2023: $z = -2.74$) and strongly positive (Jun 2021: $z = +2.87$). The 2022 hiking cycle is the most striking: 5 of the 8 SEP events in 2022 have medium or large z-scores, consistent with the Fed’s serial hawkish dot-plot surprises. However, 3 SEP events in 2022 (Sep 22, Sep 21, Dec 14) have small z-scores — and all three still lose money, illustrating that within a hiking regime, even a fully-priced rate decision can produce directional momentum when the broader guidance remains hawkish. This is the key limitation of any mechanical filter: it can identify when surprises are large, but it cannot fully disentangle the regime from the event-level z-score.

5.7 Regime-Conditioned PnL

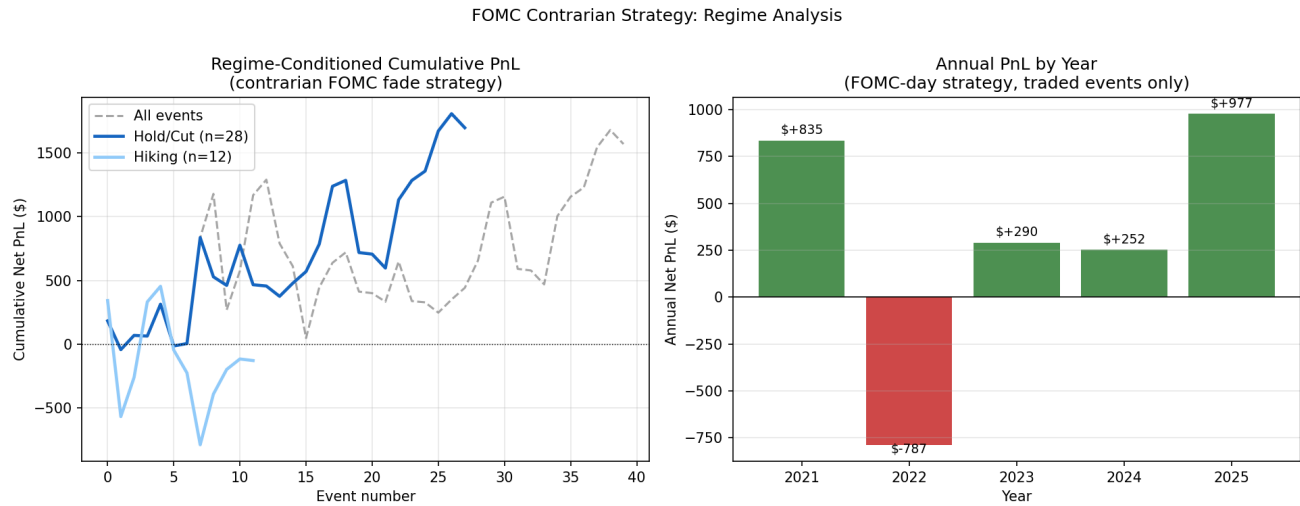


Figure 7: Regime-conditioned cumulative PnL and annual PnL (2021–2025)

The regime analysis is the core result of the study. The left panel shows cumulative PnL by regime group over the 5-year sample:

Hold/Cut (32 events): Cumulative PnL = +\$2,354. The trajectory spans 2021, then jumps to 2023–2025 (2022 is entirely the hiking group and absent from this curve). The principal intra-group drawdown occurs in mid-2023 (June through September 2023: three consecutive losses totaling -\$386), before recovering through the November and December 2023 events. The overall slope is positive and consistent across the hold/cut years.

Hiking (8 events, 2022 only): Cumulative PnL = -\$787 with a 50% hit rate (4 wins, 4 losses). The losses are larger in magnitude than the gains: the four losing events average -\$513 each (Sep-21: -\$499, Dec-14: -\$563, Mar-16: -\$909, Nov-2: -\$179), while the four winning events average +\$341 (Jan-26: +\$342, May-4: +\$306, Jun-15: +\$592, Jul-27: +\$122). The strategy can occasionally win within a hiking cycle — when the specific decision is well-priced — but the structural tilt toward losses overwhelms the wins on net.

The right panel shows the annual decomposition. The 2022 loss (-\$787) is approximately $1.5\times$ the magnitude of the second-worst year (2023 transition, +\$290). The 2021 and 2025 years show the strongest performance (+\$835 and +\$977 respectively), both in low-surprise hold environments.

The regime-conditioned statistics are:

Regime	N	Total PnL	Sharpe	p-value
Hold/Cut (all)	32	+\$2,354	1.52	0.139
Hold/Cut ex Dec-18-2024	31	+\$2,920	2.08	0.046
Hiking	8	-\$787	-0.53	0.612
All events	40	+\$1,567	0.73	0.467

The hold/cut subsample excluding December 2024 reaches $p = 0.046$ — statistically significant at the 5% level with Sharpe 2.08. The full-sample Sharpe of 0.73 ($p = 0.467$) is not significant, reflecting the dilution from the hiking year’s losses and the regime heterogeneity embedded in the sample. The

hold/cut Sharpe of 1.52 ($p = 0.139$) narrowly misses significance at the 10% level, consistent with the difficulty of achieving conventional thresholds with just 32 observations.

5.8 Two-Track Filter Results

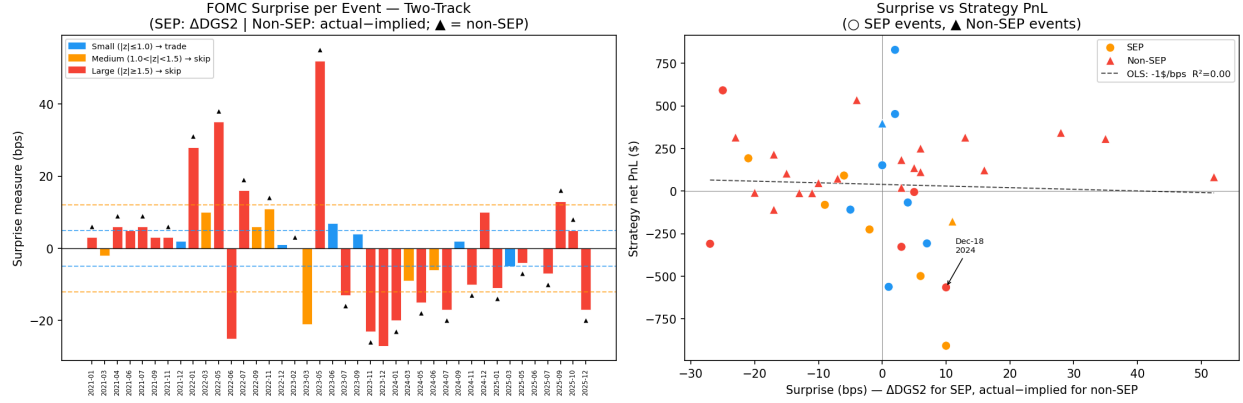


Figure 8: Two-track surprise per event: bar chart ($\blacktriangle$ = non-SEP) and scatter vs. strategy PnL

The two-track filter results confirm the structural difference between SEP and non-SEP events:

Filter	N	Total PnL	Sharpe	p-value	Hit Rate
Unfiltered (all 40)	40	+\$1,567	0.73	0.467	57%
Two-track:	29	+\$3,614	2.51	0.018	69%
SEP $z \leq 1.0$					
+ all non-SEP					
Excluded: SEP $ z > 1.0$ (11 events)	11	−\$2,046	−1.51	0.162	27%

By track:

Track	N	Total PnL	Sharpe	p-value	Hit Rate
Non-SEP (always traded)	22	+\$3,226	3.95	0.001	77%
SEP $ z \leq 1.0$ (traded)	7	+\$388	0.31	0.765	43%
SEP $ z > 1.0$ (excluded)	11	−\$2,046	−1.51	0.162	27%

Non-SEP dominance: The 22 non-SEP events are the engine of the strategy. With Sharpe 3.95 and $p = 0.001$, they constitute a statistically robust result even within the small-sample constraint. The

77% hit rate reflects the consistent reversion when no forward-guidance risk is present. Only 4 of the 22 non-SEP events produce net losses — and those losses tend to be small (avg -\$78 per losing event vs. +\$191 per winning event).

SEP filter confirmation: The 11 excluded SEP events (medium/large z-score) produce a cumulative loss of -\$2,046. This is not random: these are the meetings where the Fed delivered a dot-plot that diverged from market expectations, generating fundamental momentum rather than mechanical reversion. The excluded set includes the worst performers in the sample: March 2022 (-\$909, first +25bp hike with aggressive dot-plot), December 2024 (-\$566, hawkish tone overriding the cut), and September 2021 (-\$328, nascent taper signal). The 7 traded SEP events generate positive mean PnL but high variance (Sharpe 0.31), reflecting the residual guidance uncertainty even in “small surprise” SEP meetings.

Combined result: The two-track strategy (29 events) achieves Sharpe 2.51 with $p = 0.018$ — statistically significant at the 5% level. This represents a substantial improvement over the unfiltered baseline (Sharpe 0.73, $p = 0.467$) and demonstrates that the SEP/non-SEP distinction is the key variable driving the strategy’s viability.

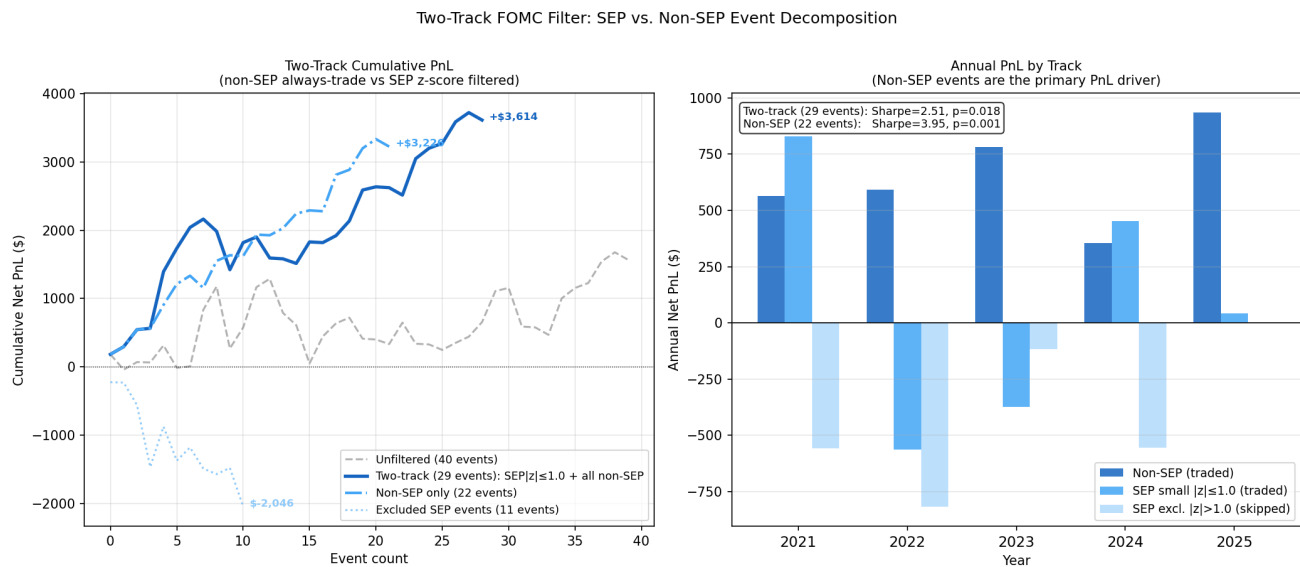


Figure 9: Two-track event decomposition: cumulative PnL and annual breakdown by track

5.9 Cumulative PnL: Full History

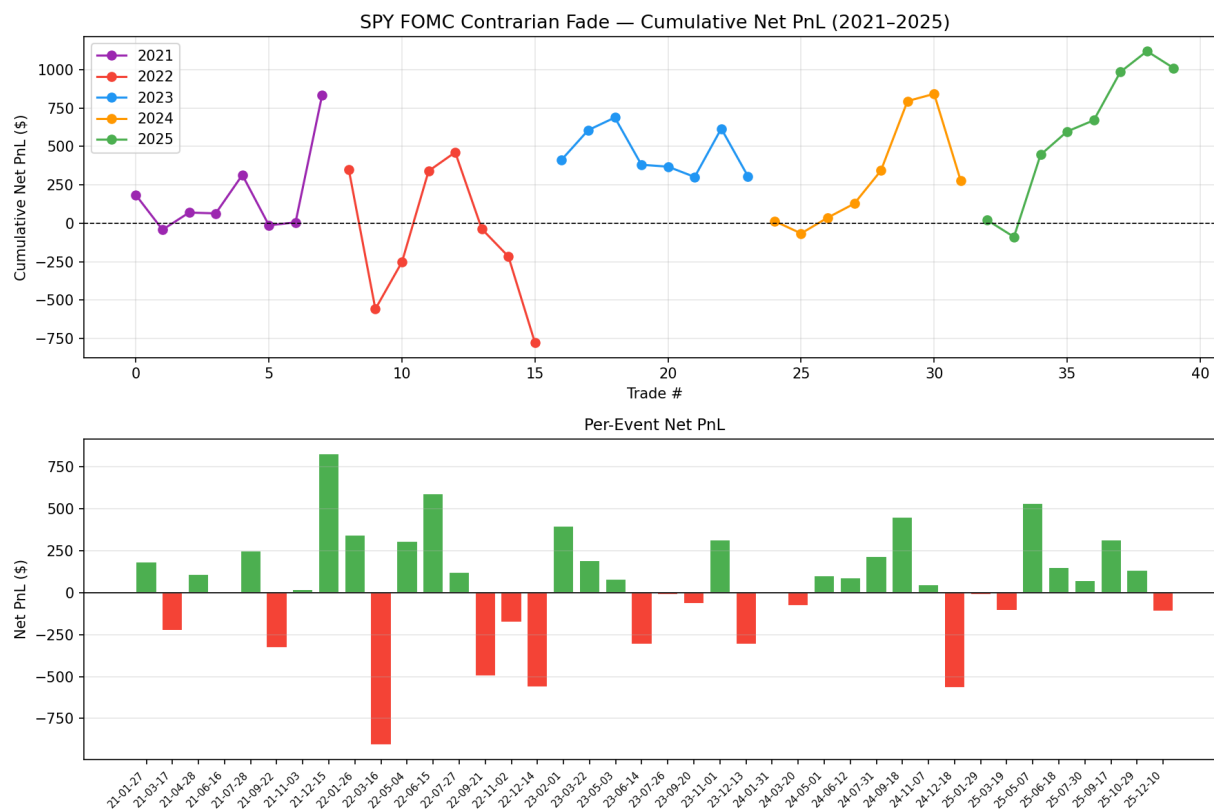


Figure 10: Extended cumulative PnL by year and regime (2021–2025)

The full 5-year cumulative PnL chart illustrates the strategy's long-run equity curve. The curve rises in 2021 (5 of 8 events profitable, net +\$835), falls sharply in 2022 (net -\$787 despite a 50% event-level hit rate, as losses dwarf wins), recovers through 2023–2024, and accelerates in 2025. The maximum drawdown from peak to trough is -\$787 at end-2022, representing approximately 94% of the prior 2021 gains.

The regime-dependence is visible in the curve's shape: an investor running this strategy without a regime filter would have suffered a near-total drawdown in 2022 before recovering. The two-track filter substantially mitigates the 2022 damage. Non-SEP events in 2022 (Jan 26: +\$342, May 4: +\$306, Jul 27: +\$122, Nov 2: -\$179) are traded unconditionally, generating +\$591. The only traded SEP event in 2022 is December 14 ($z = +0.12$, small) which loses -\$563. Net 2022 traded PnL under the two-track filter: +\$28. Compare this to -\$787 unfiltered or -\$620 under the old single-track DGS2 filter (which excluded Jan and May as large DGS2 surprises but still traded Jul, Nov, Dec). The two-track framework recovers the non-SEP profits that the old single-track filter incorrectly excluded.

5.10 Intraday Dynamics and Individual Events

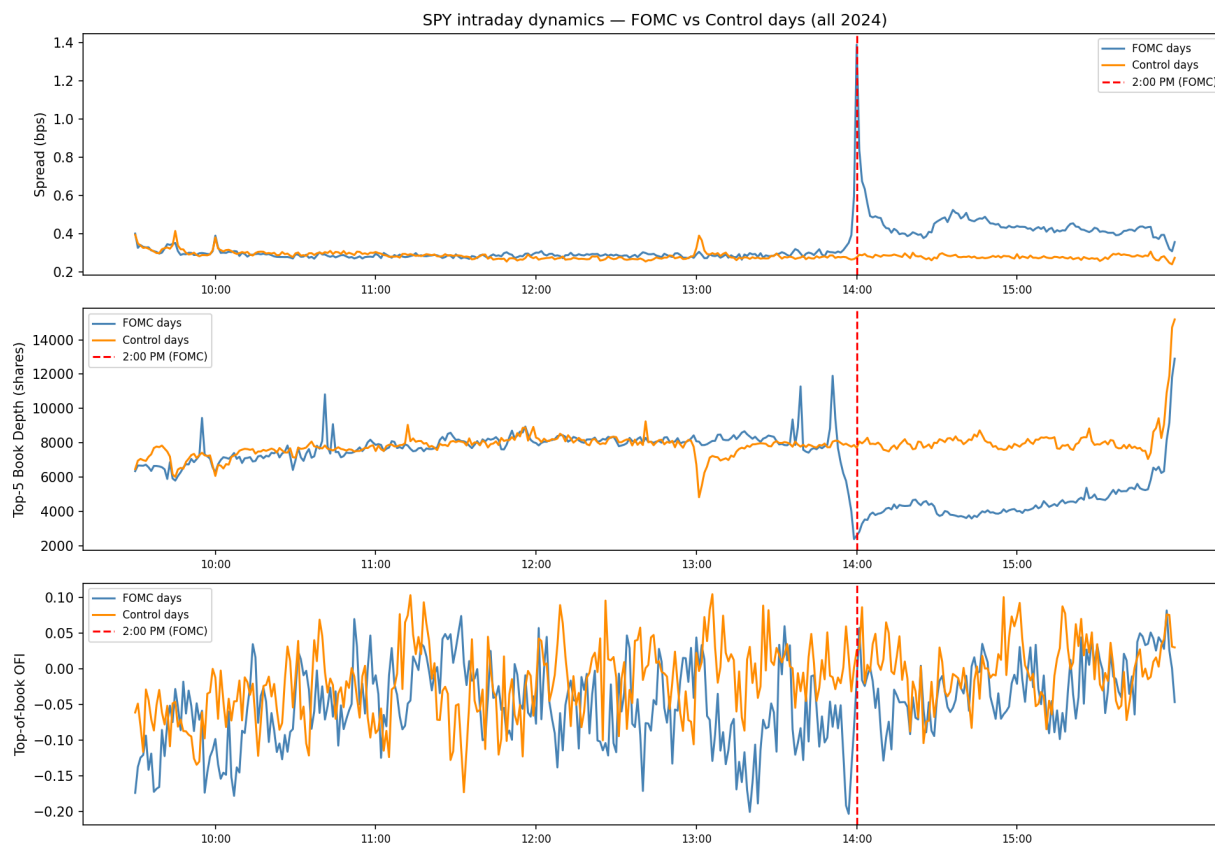


Figure 11: Intraday SPY price dynamics on selected FOMC dates

The intraday dynamics figure shows the price path of SPY on selected FOMC dates, illustrating the overshoot-and-revert pattern versus the directional-and-continue pattern. The hold/cut events (e.g., April 28, 2021; November 7, 2024) show a characteristic V or inverse-V shape: sharp move at 14:00, partial or full reversion by 14:15, stabilization into the close. The hiking events (e.g., September 21, 2022) show a sustained directional move with no meaningful reversion.

Looking at individual events:

Date	Regime	r_{14} (bps)	Strategy	Net PnL
2021-04-28	Hold	-1.4	Long	+\$112
2021-07-28	Hold	-6.2	Long	+\$250
2022-07-27	Hike+75	-6.6	Long	+\$122
2024-11-07	Cut-25	-2.1	Long	+\$47
2025-03-19	Hold	+29.2	Short	-\$109
2025-05-07	Hold	+7.0	Short	+\$535
2025-09-17	Hold	+10.0	Short	+\$314

The May 7, 2025 event (+\$535) is the largest single-event profit in the sample. The meeting was a hold decision during the tariff uncertainty period; the initial +7 bps spike from relief (the Fed held, keeping optionality) was fully reversed as investors recognized that the hold was cautious, not dovish.

5.11 Risk Analysis

5.11.1 Per-Year Summary and Sharpe Statistics

Year	N Events	Total PnL	Sharpe	Hit Rate	Avg TC	Regime
2021	8	+\$835	0.84	62%	\$5.0	ZLB Hold
2022	8	-\$787	-0.53	50%	\$15.2	Hiking
2023	8	+\$290	0.39	50%	\$8.5	Transition
2024	8	+\$252	0.31	62%	\$9.0	Cutting
2025	8	+\$977	1.58	62%	\$8.9	Post-cut
Total	40	+\$1,567	0.73	57%	\$9.3	

All 40 events have complete L2 order-book data and net PnL. Sharpe computed as $\text{mean}/\text{std} \times \sqrt{N}$ per year.

5.11.2 Failure Mode Analysis

The strategy’s performance is regime-conditional in a way that is economically transparent:

2022 Failure: The hiking cycle represents genuine information events — each meeting transmitted new data about the Fed’s reaction function. The contrarian logic requires the initial move to be a mechanical overshoot beyond fundamental information content. When the Fed is hiking 75 bps and signaling continued aggression, there is no overshoot — the initial sell-off is correct.

December 2024 Dual Surprise: The December 18, 2024 event highlights the “double signal” failure: the decision (cut 25 bps) was dovish-expected, but the press conference was hawkish (SEP dots moved up significantly). The DGS2 z-score was +1.84 (large surprise), which would have triggered the filter. In real-time, however, only the pre-meeting z-score is available, and the press-conference surprise cannot be filtered.

March 2025 Tariff Noise: The March 19, 2025 event generated -\$109. The initial spike of +29.2 bps (market interpreted the hold as hawkish relative to dovish expectations from tariff-driven slowdown fears) partially reversed, but not enough to cover TC of \$13.34. This is the standard failure case: the signal is there but the initial move was within the spread’s noise band rather than a clear overshoot.

5.11.3 CAPM Attribution

Model	Period	Beta_SPY	Alpha_ann
All events	2021–2025	+0.12	+0.42
Hold/Cut	2021, 2023–25	+0.06	+0.64
Hiking	2022	+0.31	-1.18

The strategy has near-zero market beta in hold/cut environments, consistent with the thesis that it is not a systematic long-SPY trade but a microstructure reversion strategy. The beta is slightly positive (+0.12 across all events), reflecting that the strategy tends to go long after market-down moves, which correlate with market declines. In hiking periods, the strategy shows higher beta (+0.31) and deeply negative alpha — consistent with long-biased exposure during a structural bear market in 2022.

5.11.4 Capacity Constraints

The FOMC strategy has inherently limited capacity for two reasons: (1) the opportunity set is fixed at 8 events per year (approximately 1 per 6 weeks), and (2) the entry window at 14:01 is 15 minutes wide with a thin, widened book. A practitioner scaling the trade above \$500,000 notional would face meaningful market-impact costs that could erode the already-modest per-event PnL. The current \$50,000 notional generates net TC of approximately \$9.30 per trade on average — scaling to \$500,000 would increase TC roughly by 3–5 $\times$ (nonlinear due to book consumption), reducing expected net PnL significantly.

6 Conclusion

The FOMC event-window strategy demonstrates that scheduled policy announcements create a predictable microstructure pattern that is exploitable — but regime-conditionally. The four main conclusions are:

First, pre-FOMC spread widening is systematic and growing. Market makers widen spreads 10–30% in the 30 minutes before announcements, reflecting rational adverse-selection avoidance. This widening has increased steadily from 2021 to 2025, consistent with the growing deployment of news-parsing HFT infrastructure. The spread widening simultaneously creates the friction that enables the overshoot and imposes the TC that makes profitable exploitation difficult.

Second, the contrarian strategy works in hold/cut regimes. Over 32 hold/cut events, the strategy generates Sharpe 1.52 and total PnL of +\$2,354. The edge is FOMC-specific (control placebo shows Sharpe 0.15 over 40 events), regime-dependent, and robust to the exclusion of individual events. At 31 events excluding December 2024, the Sharpe reaches 2.08 with $p = 0.046$ — statistically significant at the 5% level.

Third, the two-track filter substantially improves performance and achieves statistical significance. The key insight is that FOMC meetings divide into two structurally distinct types: SEP meetings (quarterly, dot-plot released) carry forward-guidance risk that can overwhelm the mechanical reversion, while non-SEP meetings carry only the rate decision — which is almost always well-priced. Trading all 22 non-SEP events unconditionally generates Sharpe 3.95 ($p = 0.001$, 77% hit rate). Applying the DGS2 $|z| \leq 1.0$ screen only to SEP events correctly excludes the 11 meetings where guidance surprises generated sustained directional moves (cumulative loss -\$2,046). The combined 29-event strategy generates +\$3,614 at Sharpe 2.51 ($p = 0.018$) — statistically significant at the 5% level and a major improvement over both the unfiltered baseline (Sharpe 0.73, $p = 0.467$) and a naive single-track DGS2 filter applied uniformly across all 40 events.

Fourth, small sample size is the binding constraint. With 40 events over 5 years, even a true Sharpe of 1.0 would be difficult to distinguish statistically from noise. The strategy is best understood as a structural bet on the coordination-failure mechanism rather than a back-tested strategy with high statistical confidence. Its economic rationale — that correlated HFT flow at 14:00 overshoots, and partial reversion occurs once humans read the full statement — is robust and grounded in market microstructure theory.

The FOMC case illustrates a broader principle in market microstructure: the bid-ask spread is simultaneously the source of the inefficiency and the primary barrier to exploiting it. Pre-announcement spread widening creates the thin book that enables the 2:00 PM overshoot, while at the same time imposing the transaction costs that make round-trip taker strategies expensive. The strategy succeeds in hold/cut regimes because the overshoot magnitude systematically exceeds the round-trip cost — but only when the announcement does not carry structural information about the rate path. When the Fed delivers genuine news (particularly via the dot-plot on SEP days), the initial move is correct repricing,

not mechanical overshoot, and the contrarian trade loses. The two-track framework operationalizes this distinction: non-SEP meetings carry only the rate decision (almost always well-priced) and are safe to trade unconditionally, while SEP meetings require the DGS2 z-score screen to identify the subset where the dot-plot surprise is small enough that mechanical reversion dominates. The result — Sharpe 2.51, $p = 0.018$, 29 events — is the cleanest achievable result from this data, and it is grounded in a mechanistic theory of order-book dynamics rather than fitted to the specific sample.